Module B1 — Development of Practical Skills

1.1 Planning Experiments

Planning is the first and arguably most important stage of any biological investigation. A well-planned experiment will produce reliable, valid data that can be used to draw meaningful conclusions. Poor planning leads to ambiguous results and wasted time.

Hypothesis: A testable prediction about the relationship between variables, based on existing biological knowledge or observations.
Null hypothesis: A statement that there is no significant difference or relationship between the variables being investigated.

Identifying Variables

Every investigation involves three types of variable that must be clearly identified at the planning stage:

- **Independent variable:** The factor that is deliberately changed or manipulated by the experimenter
- **Dependent variable:** The factor that is measured or observed as a result of changing the independent variable
- **Control variables:** All other factors that must be kept constant to ensure a fair test and that any change in the dependent variable is due only to the independent variable

When planning a biological experiment, you must consider the range (minimum to maximum values) and intervals of your independent variable. The range should be wide enough to show a clear trend, while intervals should be close enough to identify any anomalies or changes in the pattern.

Selecting Appropriate Apparatus and Techniques

The choice of apparatus depends on what you are measuring, the degree of precision required, and the biological system involved. Consider the following:

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- Measuring cylinders versus pipettes for liquid volumes (pipettes are more precise)
- Digital balances for mass measurements (resolution typically 0.01 g or 0.001 g)
- Colorimeters for measuring colour changes quantitatively rather than by eye
- Data loggers for continuous measurements over time (e.g., temperature, pH, light intensity)
- Microscopes with appropriate magnification and staining techniques for cellular observations

Exam Tip: When describing your method, always specify the exact concentrations, volumes, temperatures, and time intervals you would use. Vague statements like "add some enzyme" will not gain marks. State "add 5 cm3 of 1% amylase solution using a graduated pipette."

Risk Assessment

All biological practicals require a risk assessment. You must identify hazards (anything that could cause harm), assess the risk (likelihood and severity of harm), and describe control measures (steps to reduce risk). Common biological hazards include:

- Corrosive or irritant chemicals (acids, alkalis, stains)
- Biohazards (microorganisms, body fluids)
- Sharp instruments (scalpels, glass slides)
- Electrical equipment near water
- Allergenic substances (pollen, latex)

Designing Controls

A control experiment is set up identically to the test experiment but without the independent variable being applied. This allows you to confirm that any changes observed are genuinely due to the independent variable and not some other factor. In biology, negative controls (where no effect is expected) and positive controls (where an effect is known to occur) are both valuable.

1.2 Implementing Experiments

Following Procedures Accurately

Implementation involves carrying out the planned method with precision and care. In biology, many variables are difficult to control perfectly (e.g., biological variation between organisms), so careful technique is essential.

Key aspects of good implementation include:

- Following the planned method exactly, recording any deviations
- Taking measurements at the correct times
- Using apparatus correctly (e.g., reading a meniscus at eye level, zeroing a balance before use)
- Working aseptically when handling microorganisms
- Handling biological specimens ethically and safely

Recording Data

Raw data should be recorded in a clearly designed results table with appropriate headings, units, and number of decimal places consistent with the resolution of the measuring instrument. Tables should include space for repeat readings and calculated means.

Feature	Requirement
Column headings	Quantity / unit (e.g., Temperature / degrees C)
Independent variable	First column, in a logical order (ascending or descending)
Dependent variable	Subsequent columns, with repeats and mean
Decimal places	Consistent, matching instrument resolution
Units	In header only, not repeated in every cell

Aseptic Technique

When working with microorganisms, aseptic technique prevents contamination of cultures and protects the experimenter. Key steps include sterilising equipment using an autoclave or flaming, keeping Petri dishes closed when not in use, working near a Bunsen flame to create an updraft, and disposing of cultures safely by autoclaving before disposal.

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Aseptic technique: A set of procedures used to prevent contamination by unwanted microorganisms during microbiological work.

1.3 Analysis of Results

Processing Data

Raw data must be processed to reveal patterns and relationships. Common processing techniques in biology include:

- Calculating means from repeat readings (excluding anomalies)
- Calculating rates (e.g., rate = 1/time, or change in quantity per unit time)
- Calculating percentage change: ((final - initial) / initial) x 100
- Using ratios and proportions
- Calculating standard deviation to assess spread of data

Standard deviation: s = sqrt(sum(x - x_mean)^2 / (n - 1))

Standard deviation quantifies the spread of data around the mean. A small standard deviation indicates data points are clustered close to the mean, suggesting high precision and reliable results. A large standard deviation indicates wide spread, suggesting less consistency. In a normal distribution, approximately 68% of values fall within one standard deviation of the mean, and about 95% within two standard deviations.

Error bars on graphs represent the variability of data, typically showing either the range, standard deviation, or standard error. If error bars of two data sets overlap, the difference between them is likely not statistically significant. Non-overlapping error bars suggest a significant difference (though a statistical test is needed to confirm).

Statistical Tests

Statistical tests determine whether observed differences or correlations are significant (unlikely to be due to chance alone). The significance level used is typically p = 0.05 (5%), meaning there is less than a 5% probability the result occurred by chance.

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Test	When to Use	Data Requirements
Student's t-test	Comparing means of two groups	Continuous, normally distributed data
Chi-squared	Comparing observed and expected frequencies	Categoric/frequency data; expected values ≥5
Spearman's rank	Testing for correlation between two variables	Ordinal or continuous data; not normally distributed

Exam Tip: When asked to suggest an appropriate statistical test, first identify what you are testing (difference between groups or correlation?), then check the data type (continuous or categoric?). State the null hypothesis, calculate the test statistic, compare with the critical value at $p = 0.05$, and state whether you reject or accept the null hypothesis in the context of the investigation.

Graphical Presentation

Graphs are essential for identifying trends, patterns, and anomalies. The type of graph depends on the type of data:

- **Line graphs:** Used when both variables are continuous. Plot individual points and draw a line of best fit (which may be curved).
- **Bar charts:** Used when the independent variable is categoric (discrete groups).
- **Histograms:** Used for continuous data grouped into classes (e.g., frequency distributions).
- **Scatter graphs:** Used to show correlation between two continuous variables.

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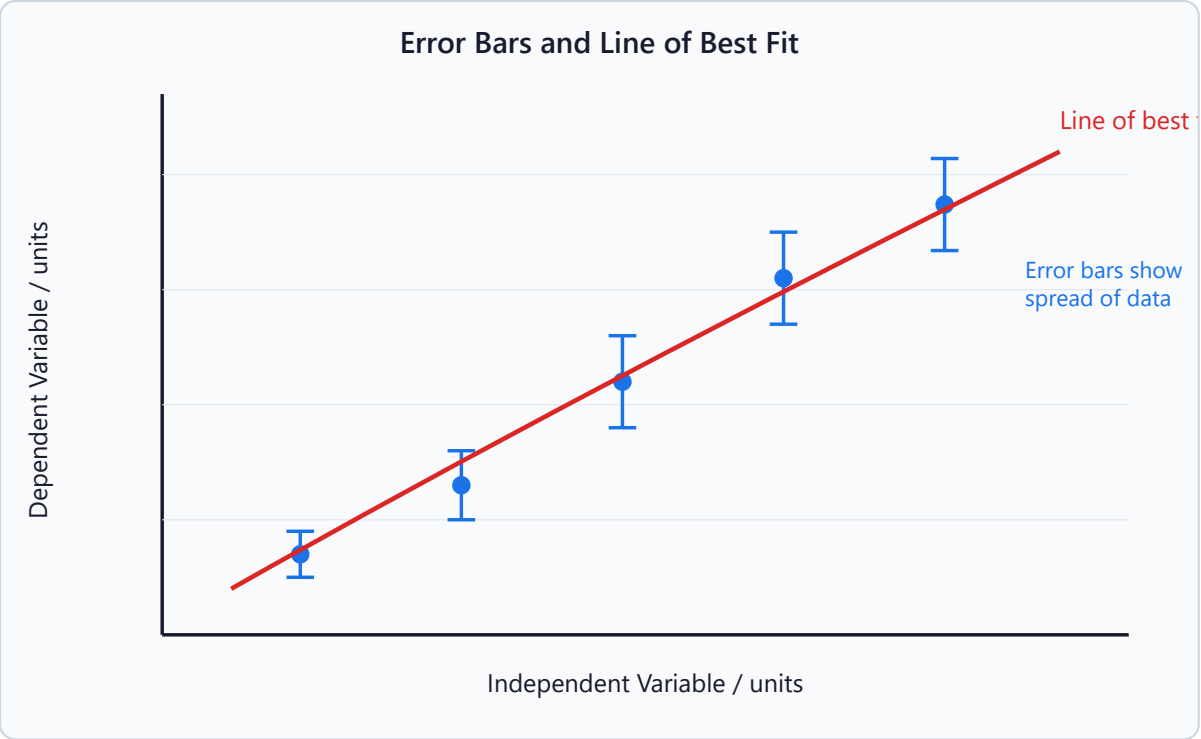


Exam Tip: Always label axes fully (quantity and unit), use a sensible scale that fills at least half the graph paper, plot points accurately with small crosses, and draw a smooth line of best fit rather than dot-to-dot connections.

Statistical Tests

Statistical tests allow you to determine whether observed differences or correlations are statistically significant (unlikely to be due to chance). The choice of test depends on the type of data and what you are testing:

Test	Used For	Data Type
Chi-squared	Difference between observed and expected frequencies	Categoric
Student's t-test	Difference between two means	Continuous, normally distributed
Spearman's rank	Correlation between two variables	Ordinal or continuous

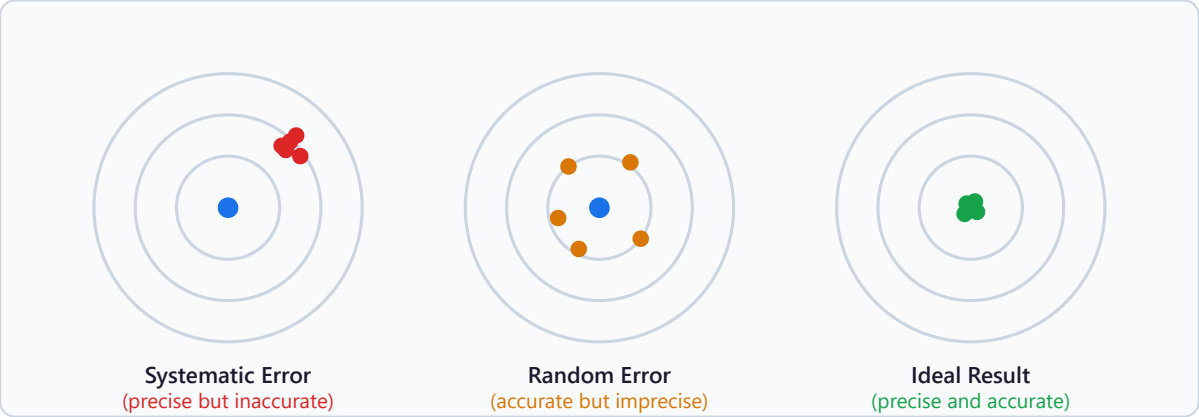
In all cases, the calculated test statistic is compared to a critical value at a chosen significance level (usually $p = 0.05$, meaning there is less than a 5% probability that the result is due to chance). If the calculated value exceeds the critical value, the null hypothesis is rejected and the result is considered statistically significant.

1.4 Evaluation

Assessing Accuracy and Precision

Accuracy: How close a measured value is to the true value.
Precision: How close repeated measurements are to each other (reproducibility).
Validity: Whether the experiment actually measures what it claims to measure and whether the control of variables is adequate.

It is possible for results to be precise but inaccurate (consistent but systematically wrong), or accurate on average but imprecise (scattered around the true value). The ideal is both high accuracy and high precision.



Identifying Sources of Error

Errors in biological experiments can be categorised as:

- **Systematic errors:** Cause all readings to be shifted in the same direction by the same amount (e.g., a balance that is not zeroed, a thermometer with an offset). These affect accuracy but not precision.
- **Random errors:** Cause readings to vary unpredictably above and below the true value (e.g., human reaction time, biological variability between organisms). These affect precision and can be reduced by taking more repeats and calculating means.

Evaluating Methods

A thorough evaluation should consider:

- Were all control variables adequately controlled? In biology, complete control is often difficult due to inherent biological variation.
- Was the sample size large enough to be representative?
- Was the range of the independent variable sufficient to identify the full pattern?
- Were there any anomalous results, and can these be explained?
- Could the method be improved to increase accuracy, precision, or validity?
- Are the conclusions supported by the data, or are there alternative explanations?

Exam Tip: When evaluating, do not simply list problems. For each limitation, suggest a specific, practical improvement. For example, "Biological variation between leaves could be reduced by using leaf discs cut with a cork borer from the same plant, ensuring uniform size and similar chlorophyll content."

Extending Investigations

After evaluating, you should be able to suggest how the investigation could be extended or modified. This might include testing additional values of the independent variable, investigating a different independent variable, using more sophisticated equipment, or increasing the sample size. You should also consider whether your conclusions can be generalised to other contexts or whether they are limited to the specific conditions tested.

1.5 Communication and Referencing

Scientific Reports

Biological investigations are communicated using a standard format that includes a title, abstract, introduction (with relevant background biology and hypothesis), method, results, discussion, conclusion, and references. Clear, precise scientific language should be used throughout, avoiding subjective or emotive terms.

Citing Sources

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When referencing published work, you should use a consistent referencing system (such as Harvard or Vancouver) and ensure all sources cited in the text appear in the reference list. This acknowledges the work of others and allows readers to locate original sources.

Peer review is the process by which scientific work is evaluated by other experts in the field before publication. This helps to identify errors, assess the validity of conclusions, and maintain the quality and credibility of published research.

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